# मेडफ्यूजन एआई (MedFusion AI)

एआई-सहायता प्राप्त व्यक्तिगत स्वास्थ्य सारांश रिपोर्ट

Institutional Ref: MF-2026-4HKG3J | Accredited Automated Analysis | A4 Clinical Standard

CONFIDENTIAL REPORT
For Patient & Physician Review

## PATIENT IDENTIFICATION

Patient Name : Test Patient
Patient ID : PID-830490
Email Address : test@example.com

## EXAMINATION & FILE DETAILS

Study Type : Cutaneous Lesion / Mole Analysis (ABCD Criteria)
Source File : sample_skin_lesion.jpg
Date & Time : 11 अक्टू॰ 2026 at 04:43 PM

**समीक्षा अनुशंसित**

Findings identified outside normal limits. Qualified medical clinician review is advised.

## EXECUTIVE CLINICAL SUMMARY

- Asymmetry score: 0.2/2.0; Border contour score: 0.5/8.0.
- Color analysis identified 3 pigment shades: Dark brown / black, Light brown / tan, Red / erythematous hue.
- Estimated major lesion diameter: 10.4 mm.

NOTABLE OBSERVATIONS REQUIRING REVIEW:

- Multiple pigment shades detected (3 colors)
- Lesion diameter exceeds 6.0 mm threshold (10.4 mm)

## SOURCE IMAGE CAPTURE



Analyzed Capture: sample_skin_lesion.jpg

## PRELIMINARY AI CLINICAL IMPRESSION

Dermoscopic evaluation identified notable structural criteria (Multiple pigment shades detected (3 colors), Lesion diameter exceeds 6.0 mm threshold (10.4 mm)). In-person clinical dermatoscopy and total-body skin examination by a qualified dermatologist are recommended.

**महत्वपूर्ण चिकित्सा अस्वीकरण**

मेडफ्यूजन एआई व्यक्तिगत स्वास्थ्य सूचनात्मक विश्लेषण प्रदान करता है। यह चिकित्सा निदान प्रदान नहीं करता, उपचार निर्धारित नहीं करता, या डॉक्टर की जगह नहीं लेता। स्वास्थ्य निर्णयों के लिए हमेशा योग्य डॉक्टर से परामर्श लें।

## DETAILED CLINICAL ANALYSIS — EXAMINATION FINDINGS

Ref: MF-2026-4HKG3J | Study: Cutaneous Lesion / Mole Analysis (ABCD Criteria)

The following structured table enumerates parameters, anatomical regions, and biomarker values evaluated by the analysis pipeline.

| PARAMETER / REGION | OBSERVED RESULT / MEASUREMENT | REFERENCE CRITERIA | STATUS |
|---|---|---|---|
| A - Asymmetry Index | 0.2 / 2.0 (Axis ratio: 1.12) | < 1.0 (Bilateral axial symmetry) | Within Limits |
| B - Border Irregularity | 0.5 / 8.0 (Circularity: 1.12) | < 3.0 (Smooth, circumscribed edge) | Within Limits |
| C - Color Variegation | 3 shades (Dark brown / black, Light brown / tan, Red / erythematous hue) | 1 - 2 uniform pigment shades | Review Required |
| D - Estimated Diameter | 10.4 mm (232 px) | < 6.0 mm (Pencil eraser benchmark) | Review Required |
| Surface Pattern & Texture | Homogeneous reticular network | Regular pigment network | Within Limits |
| Image Technical Quality | Adequate | Sharp focus, even illumination | Adequate |

### TECHNICAL QUALITY METRICS & MEASUREMENT SPECIFICATIONS

Image Dimensions: 600 × 600 px (Aspect Ratio: 1)
Sharpness Variance (Laplacian): 0.4 | Contrast Standard Deviation: 48.8
Mean Luminance Level: 205.8 / 255 | Overall Technical Grade: Adequate

Quality Note: Significant image blur detected (sharpness < 15). Findings should be interpreted with caution.

Limitations: Automated quality scoring reflects 2D sensor capture and does not guarantee complete diagnostic visibility of microscopic features.

### NUMERICAL MEASUREMENTS & ALGORITHMIC INDICES

| | |
|---|---|
| ASYMMETRY SCORE: | 0.2 |
| BORDER SCORE: | 0.5 |
| COLOR COUNT: | 3 |
| ESTIMATED DIAMETER MM: | 10.4 |

## 1. EVIDENCE-BASED SYNTHESIS & UNCERTAINTY ANALYSIS

Dermoscopic evaluation identified notable structural criteria (Multiple pigment shades detected (3 colors), Lesion diameter exceeds 6.0 mm threshold (10.4 mm)). In-person clinical dermatoscopy and total-body skin examination by a qualified dermatologist are recommended.

Uncertainty Context: Visual photography cannot replace histological pathology or biopsy. Dermatoscope illumination artifacts and magnification scaling can impact measurements.

## 2. CLINICAL NEXT STEPS & FOLLOW-UP TIMELINE

1. Schedule evaluation with a dermatologist for dermoscopy and dermpath review.
2. Perform monthly self-monitoring for any changes in size, shape, or color (ABCDE evolution).
3. Protect skin from UV exposure with broad-spectrum SPF 30+ sunscreen.

## 3. RED-FLAG WARNING SIGNS (SEEK IMMEDIATE EMERGENCY CARE)

- Severe or acute chest pain, sudden difficulty breathing, coughing up blood (hemoptysis).
- High fever (> 102°F / 38.9°C), severe dizziness, confusion, or sudden altered mental status.
- Rapidly growing, bleeding, or ulcerating skin lesion with jagged asymmetric boundaries.
- Extreme lethargy, pale clammy skin, or signs of severe dehydration / sepsis.

INSTITUTIONAL MEDICAL DISCLAIMER

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VERIFIED BY ENGINE

MedFusion AI Pipeline v2.4
SHA-256 Checksum: Verified
Signed: 11 अक्टू॰ 2026